Business Innovation Plan: AI-Powered Mental Health Care

Safe and Accessible Early-Stage Support Through a Hybrid AI-Referral Model

Summative Assessment 1

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Reader's Guide

Purpose of this Guide:

This Business Innovation Plan follows the progression that starts from a challenge question through the exploration of the challenge, to a validated solution. Each section builds on the section before it. The following table provides summaries of each section and where to locate it in this report document.

This Report follows five phases:

(1) Understanding the challenge and the competitive market

(2) Generating and selecting ideas and innovation approach

(3) Validating the selected approach with simulated users

(4) Analyzing impact, risks, and competition

(5) Identifying tools and next steps

References:

All academic, technology, course, and industry references are cited numerically throughout the document. The full citation list is available on pages 19 and 20, organized into three groups: acadmic and industry references (1 - 9_, course lecture references (L1 - L7), technology references (T1 - T7), company benchmark references (C1 - C3), and additional reading for other sources used during the exploratory phase.

Report Sections:

#	Section	Summary
1	**Exploration Report** Page 2 - 4	Defines the core challenge and benchmarks against existing tools on factors of positioning, strengths, limitations, and crisis handling. Includes a cross-industry case comparison and a proposed strategy for interviewing users for pattern identification.
2	**Innovation Report** Page 5 - 9	Defines the problem statment and the target users. Details the ideation and brainstormning processes applied to create six solution options that are scored using a NUF framework. Defines the chosen solution within the Innovation Matrix and Innovation Type, from which the value proposition, customer journey, MVP, and success metrics are defined.
3	**Validation Report** Page 10 - 12	Presents the Business Model Canvas and the user validation plan. Using the testing method’s simulatd results from interviews and prototype testing. Key insights are determined and prototype changes are outlined and aligned back to the MVP.
4	**Impact and Competition** Page 13	Provides an analysis on how the proposed model focuses on the two main barriers to adoption. Discusses the role of the freemium revenue model for increasing access. Outlines existing competitor and new entrant reactions to chosen platform model entering the market.
5	**Next Steps** Page 14	Provdes a table listing short- and long-term next steps to align to the long-term strategic goal of expansion.
6	**Future Challenges** Page 15	Outlines and explains four main risks and correlating mitigation approaches.
7	**Digital Tools and AI/IT Solutions** Page 16 - 17	Recommend a full technnology stack by functional need and reasoning for each digital solution chosen. Details how all the layers together support the platform goals and align to the challenge.
8	**Final Conclusions** Page 18	Summarizes the five main outcomes achieved in the business plan.

1. Exploration Report

1.1 Focus of the Challenge

The challenge explored is: How can AI-powered tools make it easier for users to get mental health care sooner, while upholding patient safety and trust?

Factors like costs, social stigma, and long therapy appointment wait times often delay people from getting the mental health support they need. At the same time, because of their scalability and immediate-support capabilities, AI-based conversational agents are becoming more common in mental health care. While these tools have a lot of potential to provide more access to care, concerns around safety, bias, and liability need to be further explored (6).

Research shows that AI chatbots can ease mild-to-moderate symptoms of anxiety and depression (2, 3). In order to increase the use of AI-powered tools, establishing trust and clearly managing their risks is critical. Therefore, exploring how well these tools work is critical to understand how to provide early care for more users and decrease the pressure on traditional mental health care.

1.2 Related Industry Benchmark Analysis

The following companies are used as benchmarks to compare different variations of digital mental health support.In order to assess market gaps, strengths, limitations, and crisis handling of current tools, a comparison is made ofa human-led, wellness-focused, and fully-AI tool.

Company	Core Product	Level of AI Integration	Positioning	Strengths	Limitations	Crisis Handling
BetterHelp (C1)	Online therapy with licensed professionals	Low, mostly human-led	Digitally accessible traditional therapy	Credible professional integration, curated care, flexible	Difficult to scale because of the human-aspect, high cost in comparison to other AI tools	Limited, licensed therapists are available but not in real-time, users redirected to crisis hotlines
Calm (C2)	Meditation, sleep, and stress support	Moderate, uses personalization algorithms	Preventative wellness	Broad methods, wellness-focused	Not clinical, it is more general wellness, does not have human-aspect of licensed clinicians, not focused on moderate or high symptoms	None, wellness tool that is not designed for clinical or crisis situations
Woebot Health (C3)	AI-powered Cognitive Behavioral Therapy (CBT) conversational support	AI core product, therapy via AI chatbot	Early intervention and accessibility	Evidence-supported symptom reduction	Concerns about long-term efficacy and the crisis-response abilities and safety	Key-word detection safety responses but no direct human escalation, redirected to crisis hotlines

1. 3 Relevant Cases

Industries that manage sensitive information are experiencing similar challenges to those of AI mental health. In telemedicine, research shows that if services are integrated with licensed professionals, clear standards, and structured oversight, patients are more accepting of the digital services (8). In fintech, studies show that trust in mobile financial services is highly dependent on the users' perception of security, transparency, and data protection (7). These cases confirm that when risks are clearly and properly managed, long-term adoption of AI-based mental health care tools requires establishing trust through transparency, defined limitations, data protection, and human supervision.

1.4 Strategy for User-Based Interviews and Research

The proposed strategy is focused on exploring the needs, across different user profiles, to better understand how AI-based health platforms impact hesitant users or those with experience in therapy.

Interviews will include 8-12 individuals from these 4 segments:

1. **University Students**,astheyaregenerallylowincomewithhigh stress
2. **Working professionals** whoneedeaseofaccess,flexibility,with moderate-to-high stress
3. **People who have been in therapy**,astheseindividualshaveabasis of comparison
4. **People who have not been in therapy but have considered it** to understand barriers and hesitations

Method for Gathering Information

Short, 30-45 minute, semi-structured interviews that are conducted virtually in an open-ended, exploratory question and discussion format.

Interview Topics and Questions

- Comfort levels of discussing mental health in a digital space:
  - How comfortable would you feel sharing your mental health struggles through an app or an online chat? What about when compared to talking to someone in person?
  - Have you ever searched for mental health information or support online? What was your experience like?
- Trust of AI in comparison to human professionals:
  - What would make you trust an AI mental health tool more? What would make you trust it less?
  - Would knowing a licensed professional reviews your sessions change how you feel about using an AI tool?
- Safety risks and barriers:
  - If you were in a moment of crisis, how confident would you feel that an AI tool could respond appropriately and get you the help you need?
  - What concerns do you have about your personal information being stored by a health app?
- Willingness to use AI as a first step, versus supplementary tool, versus alternative tool:
  - Would you be more open to trying an AI mental health tool as a first step before seeing a therapist? In combination with one? Why?
  - If therapy wait times for you were 2-3 months long, would that change your willingness to try an AI-supported tool?

Data Analysis

The analysis of the gathered information will be coded thematically into data that can help to identify patterns of trust, barriers, safety expectations, risks, and additional information across the different user segments.

2. Innovation Report

2.1 Context and Problem Statement

Element	Detail
Context	Early-stage anxiety and depression support are underserved due to the cost of therapy, social stigmas, and long therapy appointment wait-times.
Target User	Individuals who experience mild-to-moderate anxiety or depression symptoms who do not utilize therapy.
Problem Statement	Many individuals avoid early mental health services due to the barriers to entry and the concerns about safety and trust in AI tools.
Why It Matters	The severity of symptoms, and social and health care impacts, increase with late symptom intervention.
Success Measurement	60% two-week early-stage engagement with the tool, <10 minute escalation response time measures the success of the platform's functionality.

2.2 Idea Generation and Ideation Process

Ideas were generated using brainstorming and "How Might We?" questioning to consider different approaches to the problem (L3). To develop breadth of ideas and decrease idea judgement, quality was prioritized over quantity. Then, the NUF framework was used to score if the ideas were New, Useful, and Feasible (L4). The six main options below show different levels of risk, scalability, feasibility, and clinical impact across a range of AI-powered ideas.

#	Idea	New	Useful	Feasible	NUF Score	Reasoning
1	A full-AI therapist.	5	3	2	10	High scalability, difficult to regulate and identify early warning signs
2	Hybrid: AI manages the early-stages and automatically creates a referral to a licensed professional for identified high-risk cases.	4	5	4	13	Medium scalability, risk of missed warning signs
3	AI manages daily interactions that are reviewed by licensed professionals, with human intervention when necessary	3	5	3	11	Medium-high scalability,more crisis detection potential, lower symptom-identification and regulatory risk
4	AI-based wellness support through a short guided program	2	3	5	10	Low risk, limited clinical effects
5	Peer supported matching.	3	4	3	10	Informal, accessible, risk of low clinical effects and liability
6	AI mood-tracking with Alerts.	3	4	4	11	Low risk,limited clinical effects

The hybrid AI+referral model has the highest NUF score (13/15). This model balances usefulness and feasibility while providing differentiation. Although this model scored highest in “New”, it has liability and safety risks that decrease its feasibility score. Other lower-risk options are more feasible, but have insufficient clinical impact. The selected hybrid approach is a balanced model that focuses on early intervention and addresses liability, crisis-identification, AI-bias, and safety concerns (6).

2.3 Innovation Matrix

The hybrid AI+referral model follows Incremental Innovation within the existing digital mental health market (6, L4). It does not invent a new technology or market, but instead improves on an existing AI mental health tool product category. By integrating consistent risk-monitoring and escalation through referrals, this model enhances current AI tools. This decreases the risk from regulation and allows for faster pilot testing through a staged deployment, instead of fast iteration.

Per *Figure 1*, this solution is positioned in the Existing Market x Existing Technology section of the Innovation matrix (L1). The hybrid model uses proven CBT-supported methods that are delivered through an existing AI chatbot infrastructure with the added licensed professional referral layer (2).

This model follows a partially open innovation approach (1). It relies on external partners and licensed professionals by building off of the established CBT framework (2). It also aligns with HIPAA and GDPR regulations and healthcare advisors (T3, T6). Rather than being developed completely in-house, this approach uses outside expertise to ensure that the model is validated before it is scaled.

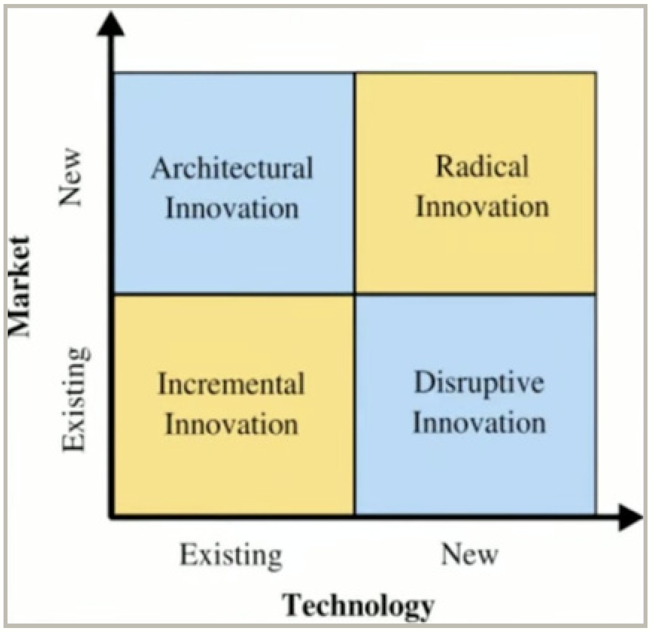

Figure 1: Innovation Matrix (L1)

2.4 Opportunity Development

Value Proposition

For people experiencing early anxiety or depression symptoms and face barriers to accessing therapy, the AI+referral solution provides immediate and evidence-backed conversational support with risk-based escalation. Unlike AI-only tools, it combines continuous risk-monitoring with licensed professional intervention. The core values include immediate accessibility, evidence-backed therapy, therapy normalization, and a clear protocol for crises.

Customer Journey

The customer journey, in *Figure 2*, maps five key user stages: Awareness, Onboarding, AI Support, Risk Monitoring, and Escalation (L2). This structure addresses the concerns about crisis-response and the feasibility of AI-powered therapy (4).

	AWARENESS	ONBOARDING	AI SUPPORT	RISK MONITORING	ESCALATION
USER ACTION	User experiences stress, anxiety, or mild symptoms	User completes short, standardized risk assessment and screening	User engages with AI chatbot for guidance and further evaluation	User continues with ongoing use of AI support services	If high-risk is detected, escalation is triggered
SYSTEM ROLE	None	AI collects baseline data and personalizes experience	AI provides CBT-based conversational support	AI analyzes user details, detects risk patterns, flags concerns	If high-risk is detected, AI refers user to licensed professional
TOUCHPOINTS	Website or App, Social media, Search	App onboarding or website interface	In-app or web chat	Back-end algorithm	Referral system and external licensed professionals
PAIN POINTS	Social stigmas, uncertainty about needing help	Privacy concerns, fear of judgment	Uncertainty about AI effectiveness	Fear of being misunderstood or misclassified	Trust in escalation need and accuracy
VALUE DELIVERED	Easily accessible starting point	Tailored support from the beginning	Immediate support that is always-available	Continuous safety monitoring	Ensures human intervention when needed

Figure 2: Customer Journey Map - Key User Stages

Prototype Development Sketches

Figure 3 shows the prototype sketches for the platform. Specifically for the risk assessment, AI-chat, and escalation alert screens a user can expect to see when using the platform.

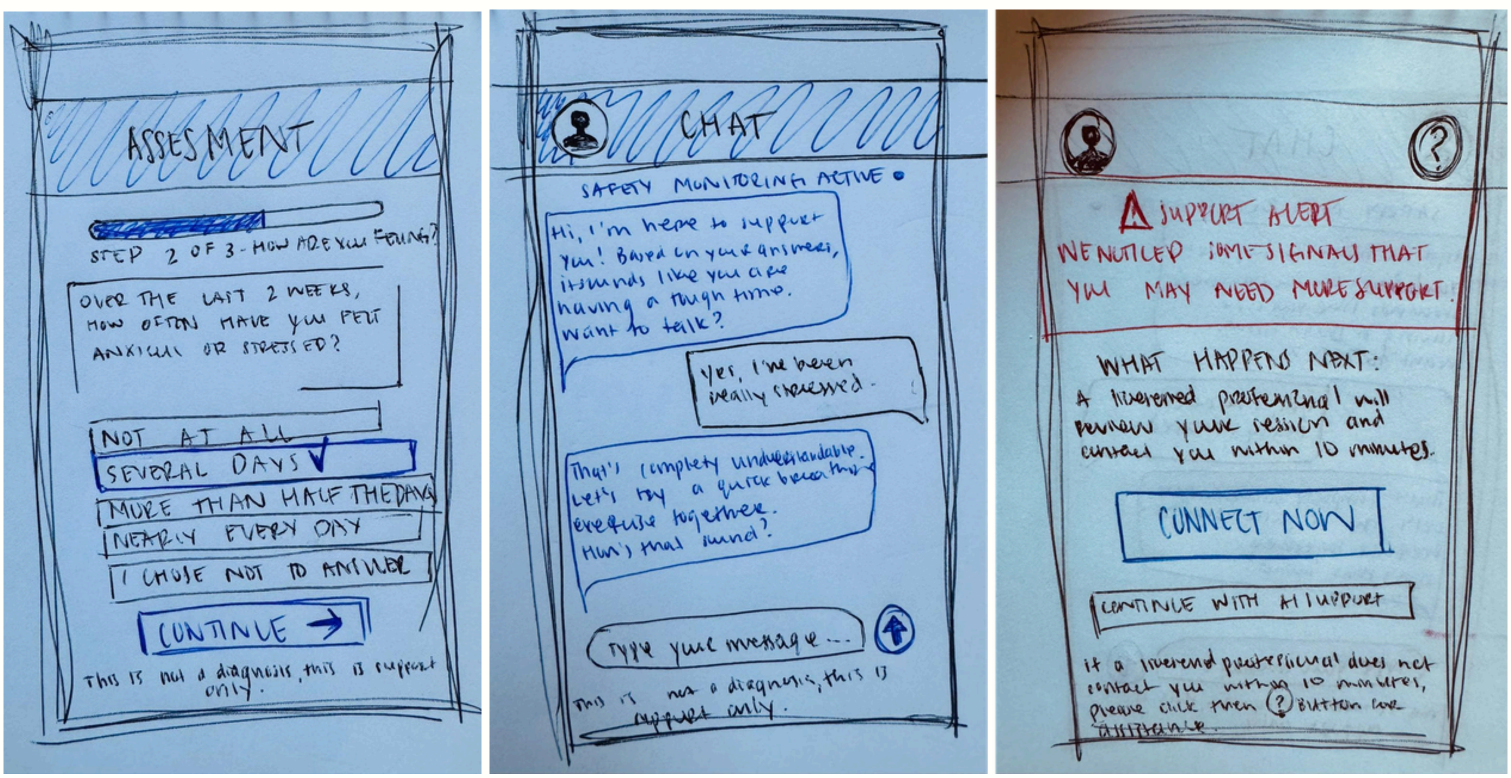

Figure 3: Prototype Sketch of Risk Assessment, AI Chat, and Escalation Alert

2.5 Minimum Viable Product (MVP) (L5)

Key Assumption: Early-stage users will consistently engage with the AI-support when safety and escalation referrals are structured.

Success Metric: 60% two-week engagement rate with <10minute escalation response time for identified high-risk cases.

The MVP Features:

- Evidence-backed CBT conversation tool
- Standardized risk-screening assessment
- Real-time risk-related keyword identification
- Clear statement, disclaimer, and safety criteria
- Direct escalation process to licensed professionals

Given the documented risks, including AI-bias and safety concerns, the MVP excludes advanced personalization or diagnosis features in order to decrease potential risks and aid in safe early-stage deployment (6).

2.6 Innovation Report Conclusion

Given that AI-tools have the potential to reduce mild to moderate symptoms, a hybrid AI+referral model providesa scalable and safe solution to mental health barriers while managing trust and liability concerns (2, 3).

3.1 Business Model Canvas

The business model in *Figure 4* addresses the market gap (L6). Existing digital mental health solutions either provide human therapy with limited scalability, or AI-only support (C1, C2, C3). Both have user safety concerns which outlines a market opportunity for the AI+referral hybrid solution.

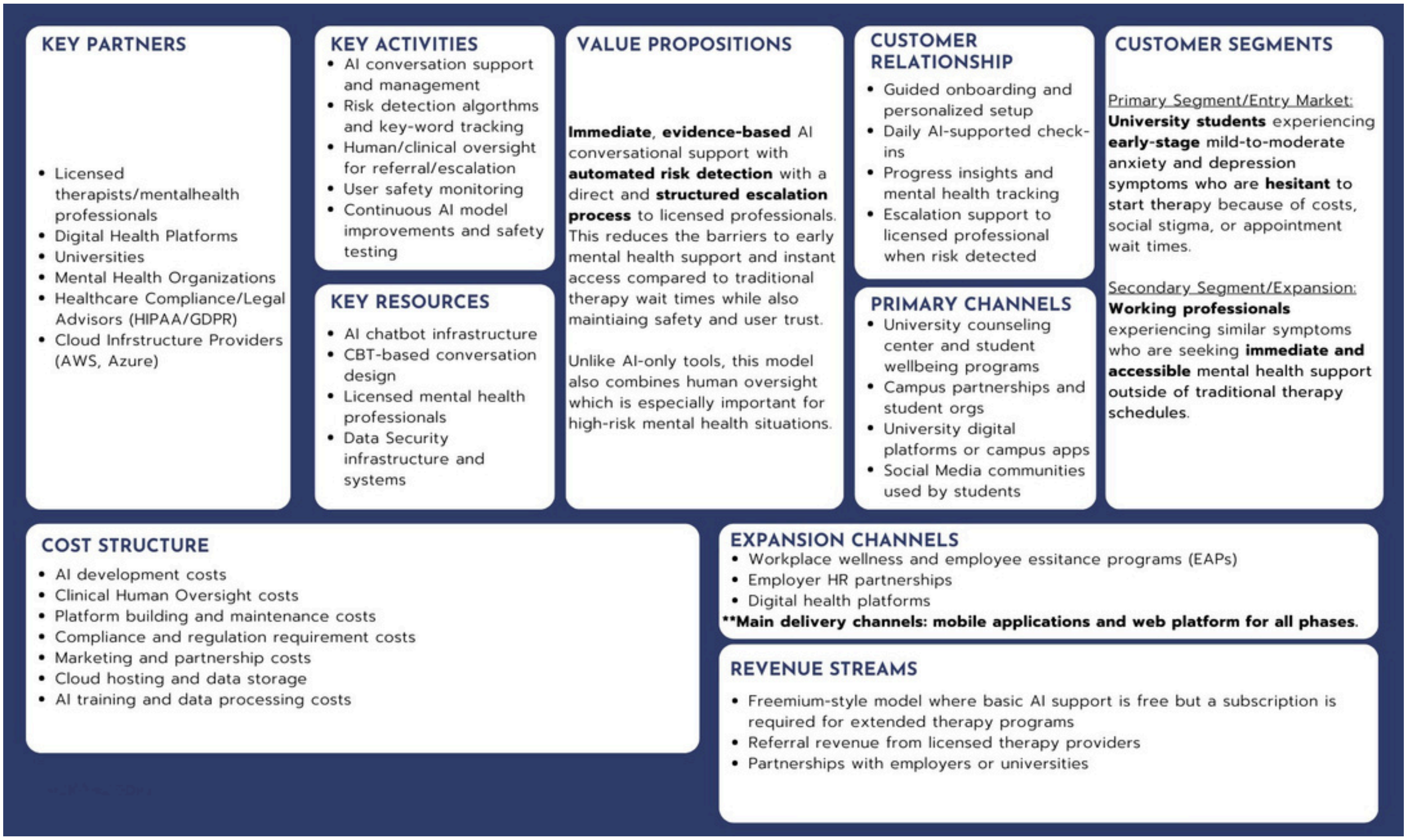

Figure 4: Business Model Canvas

3.2 User Validation Plan

The validation goal is to test the key assumption defined in the MVP: Early-stage users will consistently engage with the AI-support when safety and escalation referrals are structured.

Participants

From early research, 10 participants were recruited across four groups: university students, working professionals, people who have been in therapy, and people who have not been in therapy but have considered it. These segments allowed for comparison between experienced and hesitant users.

Method (L6)

Participants completed:

- In-person prototype walkthrough of digital mockup based on the prototype sketch of the AI conversation and escalation processes
- Scenario testing for stress, anxiety, and risk detection
- Semi-structured interviews in-person in an open-ended, exploratory format on the topics of comfort levels of discussing mental health digitally, trust in AI compared to human professionals, safety risks and barriers, and their willingness to use AI as a first step in their care

Data Analysis

The gathered information was coded thematically into data that can be analyzed to identify patterns of trust, barriers, safety expectations, risks, and additional information across the different user segments.

Success Metrics

Success criteria were aligned with the MVP goals in the Innovation Report:

- 60% two-week engagement rate with <10minute escalation response time for identified high-risk cases[1]
- User confidence in escalation safety
- Positive usability feedback from >60% of participants

[1] This is based on a study that identified that combining technology with human support has higher engagement in digital mental health intervention, but there is no universal metric for benchmark comparison as results vary across studies. Because many digital mental health tools experience a fast drop off in engagement, a 60% two-week engagement rate was chosen to indicate meaningful early adoption in the prototype testing phase (5).

3.3 Validation Results

Key Insights

Three major themes were consistent following the user testing:

1. Safety and human oversight increased trust. Participants were concerned about the AI making clinical decisions on its own until the escalation system was explained, which greatly improved trust.
2. Immediate accessibility was valued. Traditional therapy wait times were a repeating barrier so immediately access to AI support was seen as a primary benefit.
3. Users emphasized transparency about the AI's capabilities, on when escalation occurs, and on how user data would be used.

These key insights confirm the research findings that transparency and structured human oversight can improve the adoption of AI digital services.

Prototype Changes Based on Feedback (15)

Feedback	Change Implemented
Users concerned about AI misdiagnosis	Added clearer disclaimer that the tool provides support only, not a diagnosis
Users want clarity about escalation	Added a 'Safety Monitoring Active' indicator
Some users were confused during onboarding process	Simplified the onboarding process into a simple 3-step process

3.4 Validation Conclusion

The validation process confirmed a strong potential for the hybrid AI+referral model. Participants responded positively to the differentiated model's immediate accessibility and human oversight, especially when compared to traditional therapy or AI-only tools. The key insight is that, in order to push early adoption, establishing trust is more important than adding advanced AI features. Therefore, it is recommended to prioritize clarity,safety monitoring, and clear escalation protocols before adding complex personalization features.

4. Impact and Competition

4.1 How This Innovation Improves Businesses

The hybrid model directly tackles two main barriers to adoption in the AI mental health space: safety concerns and lack of trust. The added accessibility through the integration of an AI-conversation agent creates a product that is scalable. The escalation to licensed professionals adds a second layer that makes the platform clinically credible. This positions the platform to capture users who are hesitant about fully AI-tools but are not willing to wait or pay for traditional therapy.

The freemium revenue model adds more accessibility as it requires a low, or no, financial commitment. This can increase user engagement among new and hesitant users. In addition, the revenue model allows for data collection with a wider range of users to develop insights for future improvements and decision-making.

4.2 Competitor Reaction and Response

Competitors may react in one of two ways. AI-only platforms, like Woebot, may be pressured to add stronger human escalation protocols to stay competitive and align with regulations (C3). Human-led platforms, like BetterHelp, may invest in AI tools to help reduce the workload of their therapist network and improve their response times (C1). New entrants may attempt to imitate the model. Overall, when entering the market, the hybrid model's competitive advantage will lie in its speed to deployment, its clinical partnerships, and its strong credibility and trust.

5. Next Steps

5.1 Next Steps and Long-Term Strategic Goals

#	Action	Details	Timeline
1	**Pilot Test**	Structured pilot test with a university. Partnering with a credible institution allows for access to a concentrated and controlled setting. At a university, the large user base of target users allows for preliminary data collection. The pilot will provide engagement and escalation success metrics to compare against the outlined MVP targets.	Short Term
2	**Advisory Board**	Set up a clinical advisory board to manage compliance needs. Focusing on bias management through AI training through inclusive data, developing escalation protocols, aligning to regulations, and setting up clinical auditing processes.	Short Term
3	**Network Building**	Build a network of licensed professionals across time zones before the launch. The goal is to create a wide network of on-demand clinicians aligned with future growth.	Short Term
4	**AI Expectations**	Develop clear documentation with clear expectations about what the AI can and can not do, including for when human oversight is part of the process.	Short Term
5	**Certifications**	Apply for needed health data certifications to support partnerships. This includes HIPAA for managing US patient health data and GDPR for handling EU patient data (T3,T6).	Short Term
6	**Partnership - National Health Care and Insurance Providers**	Partner with national health care systems and insurance providers to expand reach and credibility. This will add credibility and help position the platform within existing and trusted healthcare systems. Also adds access to new users once the platform is part of health systems.	Long Term
7	**Expansion**	This entails adding mild-to-moderate symptom support. This step is dependent on successfully establishing and validating clinical trust.	Long Term
8	**Partnership - Workplace Programs**	Partner with HR and employee support programs to launch the platform as an employee benefit. The partnership with workplaces increases customer reach through businesses and new revenue streams.	Long Term
9	**New Market Expansion**	Enter into underserved international markets that have limited access to early symptom mental health care. This includes areas like Sub-Saharan Africa and South Asia that are considered priority areas in which technology-based mental health tools can cover gaps that traditional tools do not cover (9).	Long Term

6. Future Challenges

6.1 Key Future Challenges

Challenge	Reasoning
Regulatory and Liability Risk	AI mental health care is a highly regulated space. HIPAA, GDPR, and other AI-specific health regulations will require continuous investment in order to decrease liability risks. Specifically in the misclassification of high-risk users (4). To manage this risk, the escalation process must be transparent and all clinical validation must be documented and audited.
AI Bias and Misclassifications	Poorly trained AI can produce biased results. AI can miss high-risk signals or create unnecessary escalations if not trained properly and monitored. By using broad and all-inclusive training data and auditing results, this risk can be reduced.
Scalability of Human Oversight	The hybrid model is fully dependent on licensed professionals managing risk escalations. As the user base grows, the volume of escalations is expected to increase. If in this growth the escalation response time exceeds the target of <10 minutes, the main promise of safety may be undermined. To mitigate this risk, a wide network of licensed professional and operational coordination must be prioritized in the early stages in preparation for scaling.
Decreased Trust from Incidents	Given the human-aspect and sensitivity of the consumer market, there is a risk of decreased trust if one high-risk incident occurs. This could include a missed escalation or a data breach. To reduce this risk, incident response plans and crisis communication protocols must be built into internal processes from the beginning.

7. Digital Tools and AI/IT Solutions for Implementation

7.1 Technology Recommendation by Functional Area[2]

Core AI and Conversation Platform

Tool/Technology	Purpose	Reasoning
OpenAI API/Anthropic Claude API	Power the conversational AI chatbot	Innovative language models that are trained for nuance, safe conversations, and offer enterprise-level safety controls
Rasa	Customer conversation flow manager	Allows for full control over conversation logic, escalation triggers, and domain-specific intent recognition without the need for third-party data management
Google Dialogflow CX	Conversation design and testing	Enterprise-level conversation builder with built-in testing tools and dialogue management

Risk Detection and Safety Monitoring

Tool/Technology	Purpose	Reasoning
AWS Comprehend Medical	Real-time risk keyword detection	HIPAA-eligible service that identifies risk signals
Customer ML Model	Predictive risk scoring that is based on patterns from sessions	Allows AI training on specific data for risk identification accuracy
PagerDuty/OPSgenie	Escalating alerts for on-call licensed professionals	Industry standard incident management tools with fast alerts to align to escalation response time target

Data Privacy, Security, and Compliance

Tool/Technology	Purpose	Reasoning
AWS HealthLake/Google Cloud Healthcare API	HIPAA-compliant health data storage	Built for healthcare data with audit logging, encryption, and security controls
Okta/Auth0	Identity management and secure user authentication	Industry standard for healthcare applications that supports SSO and MFA for users and clinical professionals
OneTrust	HIPAA and GDPR compliance manager	Automates data mapping, consent management, and reporting for regulations across jurisdictions

[2] Note that all references for Section 6, Digital Tools and AI/IT Solutions for Implementation, are listed in the Technology References sections of the Reference Page for visibility purposes (T1, T2, T3, T4, T5, T6, T7).

Clinical Coordination and Professional Network

Tool/Technology	Purpose	Reasoning
Calendly/Acuity Scheduling	Automated appointment booking after escalation	Coordinates with therapists' calendars for easier and faster appointment booking after escalation
Zendesk/Salesforce Health Cloud	Case management for escalated users	Tracks escalation history, notes, and follow-up statuses to create consistent documentation
Teladoc Health API	Access to licensed mental health professionals on-demand	Established telemedicine infrastructure with global, verified, and licensed professionals

Analytics, Monitoring, and Product Improvement

Tool/Technology	Purpose	Reasoning
Mixpanel/Amplitude	User engagement and retention analysis	Tracks session frequency, drop-off points, and feature use, supports 60% two-weeks engagement metric
Grafana + Prometheus	Real-time escalation and platform performance monitoring	Open-source monitoring that tracks escalation response times, delays, and system operations
Weights and Biases/MLflow	ML model performance tracking and versioning	Monitors AI model accuracy over time, tracks bias metrics, and manages model versions

7.2 How IT and AI Use This Pathway to Succeed

The recommended technology stack is interconnected. Together, these tools drive and deliver the platform's main promises of immediate, safe, clinically supported, and trusted mental health support.

The conversation AI layers handle the front-end user experience by removing wait-time barriers. The risk detection layer addresses the users' concerns of safety and trust. The compliance and security infrastructure prepares the platform for partnerships and approval from regulators. The clinical coordination layer ensures that human oversight is present as the platform scales. And last, the analytics layer will collect and provide data for future product and improvement decisions while also training and improving the AI. Together, these layers support the platform's AI growth without decreasing the clinical oversight and accessibility factors that differentiate the AI+referral model from its competitors.

8. Final Conclusions

8.1 What Was Achieved

The Business Innovation Plan explored the challenge of improving early access to mental health support through AI-powered tools. Throughout the exploration, innovation, and validation processes, the following was understood and developed:

- There is a clear market gap where existing solutions are too expensive, slow, or lack clinical credibility and safety infrastructure.
- User research validated the assumption that when safety and escalation processes are transparent and structured, users are much more willing to use AI-powered mental health tools.
- During ideation, the hybrid AI+referral model scored the highest in NUF evaluation. It had the clearest value proposition. The model balances scalability, clinical impact, user trust, and crisis response better than the other ideas.
- The MVP has realistic and measurable targets. A 60% two-week engagement rate with <10 escalation response time are solid benchmarks for pilot testing.
- The technology stack can support the platform’s deployment without needing a full customer base.

The hybrid AI-referral model is not only a product, but it builds the foundation for scaling more accessible and trustworthy mental health care systems that is directed towards helping users before their symptoms require more support.

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Medical Daily. (n.d.). *Therapy apps vs. in-person therapy: Do digital mental health apps really work?* https://www.medicaldaily.com/therapy-apps-vs-person-therapy-do-digital-mental-health-apps-really-work-474711